

Quiz3, MAT1375 Professor Chiu

ID: _____

Name: _____

- This quiz consists of 3 sets of questions for a total of 10 points.
- You have 15 minutes to complete the quiz.
- Show all work and justify your answers.
- Wishing you success.

(5 points) 1. Consider the following graph of a function f .

Find (a) the domain of f . (b) the range of f . (c) $f(5)$. (d) $f(3)$. (e) $f(9)$.

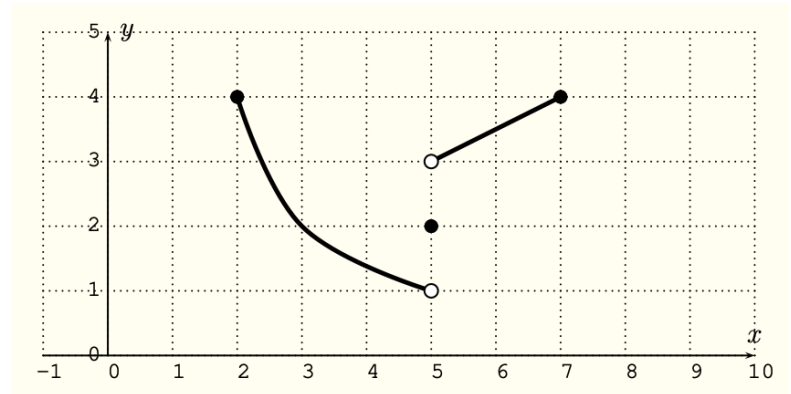
(a) $D_f = \{x | x \in [2, 7]\}$

(b) $R_f = \{y | y \in (1, 4]\}$

(c) $f(5) = 2$

(d) $f(3) = 2$

(e) $f(9)$ undefined.



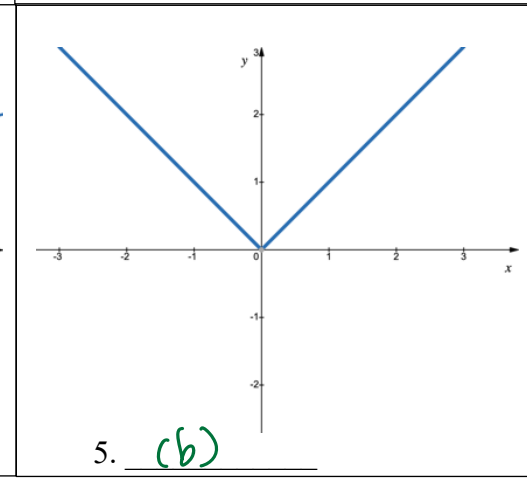
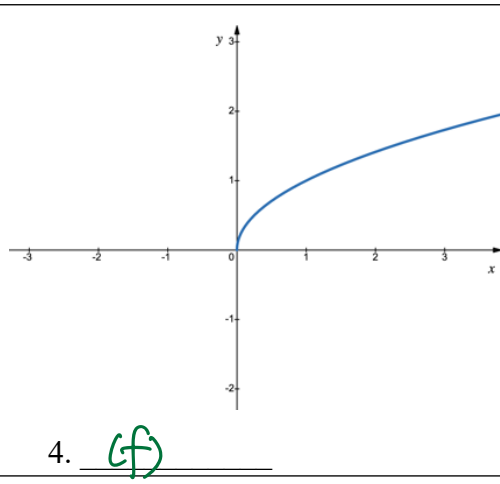
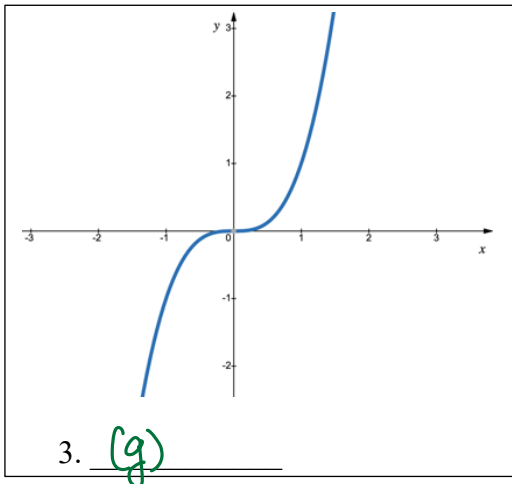
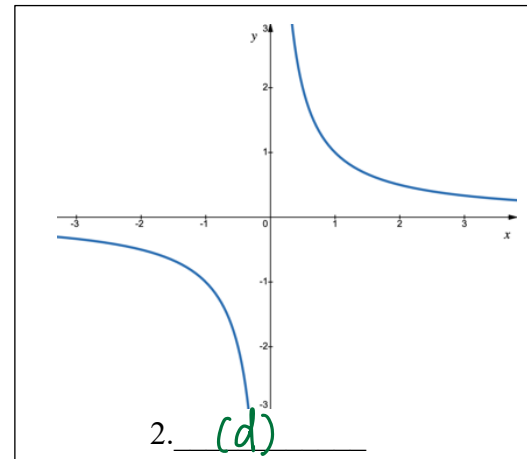
(2 points) Pair the functions (a)-(h) with the given graphs in the following question

(a) $f(x) = \text{constant}$. (e) $f(x) = x^2$.

(b) $f(x) = |x|$. (f) $f(x) = \sqrt{x}$.

(c) $f(x) = x$. (g) $f(x) = x^3$.

(d) $f(x) = \frac{1}{x}$. (h) $f(x) = \sqrt[3]{x}$.



Please turn over and finish the rest of the question.

(3 points)6. Given a function $f(x) = 3x^4 - 4x^2 + 5$. Determine if the function f is even, odd, or neither.

$$f(a) = 3a^4 - 4a^2 + 5$$

$$f(-a) = 3(-a)^4 - 4(-a)^2 + 5 = 3a^4 - 4a^2 + 5$$

Since $f(-a) = f(a) \Rightarrow f$ is even.